MILITARY SPECIFICATION

ANTENNA AT-141A/ARC

This specification has been approved by the Bureau of Aeronautics, Department of the Navy

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification establishes the requirements for antenna type AT-141A/ARC which is a streamlined blade, transmitting and receiving antenna used with airborne communication equipments operating over the frequency band of 225 to 400 megacycles per second.

1.2 Classification. The equipment covered by this specification shall be of one type and shall consist of the following:

Maj. Unit	Type Designation	Size (in)	Max. Allow. weight lbs.	Applicable Paragraphs
Antenna	AT-141A/ARC	15 x 8-7/8 x 1-1/2	2.5	3.2 and 3.3

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards drawings and publications of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Federal

QQ-P-416	Plating Cadmium (Electrodeposited)
QQ-S-571	Soldering

Military

MIL-C-71	Connectors, "N" for Radio Frequency Cables
MIL-V-173	Varnish, Moisture-and Fungus-Resistant, for the Treatment of Communications, Electronic and Associated Electrical Equipment

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SPECIFICATIONS (Continued)

Military (Continued)

MIL-E-5400	Electronic Equipment, Airborne, General Specification
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment
MIL-P-6064	Packaging of Lightweight Aircraft Accessories
MIL-C-7439	Coating, Rain Erosion Resistant and Rain Erosion Resistant with Anti-Static Surface Treatment; for Plastic Laminates
JAN-T-152	Treatment, Moisture- and Fungus-Resistant for the Treatment of Communications, Electronic and Associated Electrical Equipment, General Process for

STANDARDS

Military

MIL-STD-129	Markings of Shipments

BUREAU OF AERONAUTICS

SR-197	Nomenclature and Nameplates for Electronic and Associated Equipment
SR-190	Design Data Requirements for Contracts Covering Electronic Equipments
52A1D40-1	Antenna AT-141A/ARC, Outline and Detail Drawing

2.2 Availability of specifications, publications and drawings:

(1) When requesting specifications, publications and drawings refer to both title and number. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

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(2) Copies of this specification and applicable specifications and publications may be obtained upon application to the Commanding Officer, Naval Air Station, Johnsville, Pennsylvania, Attn: Technical Records Division.

(3) Copies of applicable Bureau of Aeronautics drawings may be obtained upon application to Bureau of Aeronautics, Navy Department, Washington 25, D.C. Attn: Electronics Division.

2.3 Precedence. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract. The contract shall have precedence over any specification.

(2) This specification. This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the Bureau of Aeronautics.

(3) Reference specifications. Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following factors shall govern:

(1) Parts and materials approved to specifications listed in MIL-E-5400 shall be given first consideration.

(2) If the contractors can demonstrate that the use of standard AN or JAN parts or materials will not fulfill the design objectives because of size, weight, performance or other reasons, material and parts shall be used which most nearly meet or exceed the performance requirements of the respective standard specification, provided that no deterioration of size, weight, performance or reliability will result in the use of such substitutions.

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(3) In all cases, the meeting of general material requirements such as non-inflammability, fungus resistance, nontoxicity, etc. and, the meeting of environmental conditions as required by specification MIL-E-5400 or MIL-T-5422 must be of prime consideration.

3.1.1 Non-standard part approval and material approval. All non-standard parts or materials (parts or materials which do not comply with MIL-E-5400) of an electrical nature or methods of manufacture therefor and techniques and procedures involving their use and application shall be submitted to the Bureau of Aeronautics for approval as required by specification MIL-E-5400.

3.1.2 Design objectives. Minimum size and weight, simplicity of operation and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives in the design of the equipment specified herein. In accordance with paragraph 3.1 the material, parts, mechanical assemblies and manufacturing processes shall be in accordance with the requirements of specification MIL-E-5400, however the use of other materials, parts and processes shall be investigated and if it appears that a substantial reduction in size and weight or improvement in simplicity or performance can be realized by their use, a request for approval of a non-standard part shall be submitted to the Bureau of Aeronautics for consideration, each request should be accompanied by complete supporting information.

3.1.3 Cadmium plating. Cadmium plating in accordance with Federal specification QQ-P-416 shall be used to insure electrical contact between brass and aluminum alloy metal.

3.1.4 Soldering. Unless otherwise specified, only non-corrosive core solder in accordance with specification QQ-S-571, class A, shall be used. Cleaning flux, such as sal amoniac, shall not be used. Alcohol may be used for cleaning after tinning and soldering. Mechanical loads shall not be imposed on soldered connections.

3.1.5 Dielectric material. The dielectric material or materials used between the stub and the sleeve of the antenna shall be chosen so that the antenna will best fulfill the requirements of this specification. If impregnated glass cloth is used in the construction of the antenna it shall be Corning Glass works type ECC-162-11 (or finer mesh) or equivalent and shall be impregnated with a suitable polyester resin having a dielectric constant of 2.75 $\pm$ .25 and a dissipation factor of no greater than .014 at 200 megacycles per second.

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3.1.6 Workmanship. All details of workmanship shall be in accordance with the best manufacturing practice for high-quality electronic equipment. Particular attention shall be given to neatness and thoroughness in the making of parts, plating, soldering and finishing.

3.2 General design requirements.

3.2.1 General. The antenna shall meet the general design and performance requirements of specification MIL-E-5400 except as modified by the following paragraphs:

3.2.2 Construction. The antenna shall have the airfoil mounting dimensions, and general features of construction as shown on Bureau of Aeronautics Drawing 52A1D40-1. The bottom portion of the antenna, which mates with the outer surface of the aircraft skin, shall be so designed as to permit essentially a high pressure electrical contact between the aircraft skin and the periphery of the base surface of the antenna, and at the same time allow for the use of a small moisture sealing gasket within the periphery of the base, without interfering with the electrical contact with the skin. Cementing of dielectric material, although desirable for sealing purposes shall not be relied upon to secure the dielectric portion of the antenna to its upper and lower metal portions. Other provisions, such as flush rivets or bolts used with grommets or pressure plates, shall be employed to insure adequate structrual strength of the antenna. The external dielectric surface shall be coated with a non-conducting, low loss dielectric material to prevent absorption of moisture or the formation of a film of moisture on the surface. In addition a Class I rain erosion coating shall be applied to the antenna in accordance with the requirements of specification MIL-C-7439.

3.2.3 Mechanical strength. The antenna shall be designed to withstand a uniformly distributed lateral ultimate design load of 9.5 pounds per square inch.

3.2.4 Mating surfaces. Electrically conduction mating surfaces shall be clean metal surfaces, free from grease, paint lacquer, anodizing or similar resistive film so as to insure negligible radio frequency impedance.

3.2.5 R-F connectors. The r-f connector included as part of the antenna shall be designed to accommodate plug UG-21/U in accordance with MIL-C-71.

3.2.6 Safetying. All internally and externally threaded parts shall be positively safetied by the use of lock washers, lock nuts, or other suitable methods.

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3.2.7 Weight. The antenna shall be of the lightest practicable weight consistent with the other requirements of this specification and in no event shall the installed weight of the antenna exceed 2.5 pounds. Installed weight shall include the complete antenna assembly with nameplate and receptacle, but shall not include cabling, external to the antenna, plugs, mounting nuts or bolts, or any mounting bracket or fastening device which is not an integral part of the antenna assembly.

3.2.8 Tropicalization. Antennas covered by this specification shall be tropicalized in accordance with specification JAN-T-152 and MIL-C-173.

3.2.9 Interchangeability. The interchangeability requirements as specified in specification MIL-E-5400 shall be extended to require interchangeability of antennas with the latest model of any previously produced antennas. Such interchangeability shall be measured against a model, manufacturing drawings or other technical information as provided by the Bureau of Aeronautics for that purpose. In the event that requirements of MIL-E-5400 conflict with the requirements for interchangeability the interchangeability requirements shall govern.

3.2.10 Collection of liquids. The antenna shall be designed and constructed such that when mounted vertically either on the top or bottom surface of an aircraft structure no water or other liquids can collect in any portion of the antenna that could ultimately result in failure or deterioration of performance of the antenna or rupture due to freezing.

3.3 Performance requirements.

3.3.1 Performance. The antenna shall be so designed that when mounted in the center of a large horizontal ground plane it shall provide vertically polarized radiation of essentially uniform intensity over 360 degrees of azimuth. The antenna shall satisfy the performance requirements which are outlined in Section 4 when subjected to the following tests:

(a) Mechanical Stress Test
(b) Insulation Test
(c) Radiation Test
(d) Voltage Standing Wave Ratio Test

3.4 Identification of product.

3.4.1 Nameplates and nomenclature. Nameplate and nomenclature approval and assignment shall be granted by the Bureau of Aeronautics only upon compliance with the applicable requirements of specification SR-197.

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3.4.2 Use on AN or MIL designations. AN or MIL designations shall not be applied to a product except for preproduction test samples, nor referred to in correspondence, sales matter or otherwise, until notification has been received from the Bureau of Aeronautics that the product has been approved. When the antenna is procured by a prime contractor through subcontract, purchase order, or otherwise (not including Government furnished) the prime contractor shall be responsible for compliance with the requirements of paragraph 3.4.1 and approval, when granted will extend only to that one antenna source or to that one prime antenna contract.

4. METHODS OF SAMPLING, INSPECTION AND TESTS

4.1 Classification of tests. The inspection and testing of antennas shall be classified as follows:

(a) Contractor tests. Contractor tests are those tests conducted by the contractor on an antenna to determine that the antenna complies to the best of his knowledge and belief with all applicable requirements.

(b) Preproduction tests. Preproduction tests are conducted by the procuring agency, after the award of the contract, on samples which are representative of the production antennas to determine that the antenna meet all the requirements of this specification.

(c) Inspection tests. Inspection tests are those tests accomplished on the antennas submitted for acceptance under the contract.

4.2 Contractor tests. The contractor shall conduct tests on one or more sample antennas as necessary to determine that the design of the antennas as proposed by the contractor will meet the requirements of this specification. Contractor tests shall be conducted in accordance with the approved preproduction test procedure. The data obtained by the contractor in conducting these tests shall be included with the design data submitted with the preproduction test antennas. The Inspector of Naval Material and the Bureau of Aeronautics shall be advised when tests are to be conducted so that the Bureau of Aeronautics may designate a representative to witness or supervise the test when so desired. When tests are supervised the Bureau of Aeronautics may decide to omit certain portions of the preproduction tests which will duplicate the contractor tests; however, in conducting preproduction tests the Bureau of Aeronautics may elect to repeat any test previously conducted by the contractor, when deemed necessary. (Contractors not having laboratory facilities to satisfactorily conduct all tests shall either obtain the services of a commercial testing laboratory or receive written approval from the Bureau of Aeronautics to omit that portion of the tests.)

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4.3 Preproduction test.

4.3.1 Sampling instructions. Preproduction test samples shall consist of three antennas representative of the production antennas to be supplied under the contract. Preproduction tests shall be conducted at laboratory designated by the Bureau of Aeronautics. Samples shall be plainly identified by securely attached durable tags marked with the information specified below, and forwarded with design data to the laboratory designated in the letter of authorization from the Bureau of Aeronautics.

Sample for preproduction test
Antenna AT-141A/ARC
Name of manufacturer
Submitted by (name) (date) for qualification in accordance with requirements of specification MIL-A-7879A(Aer) under authorization (reference authorizing letter)

4.3.2 Scope of tests. Preproduction tests shall include any tests deemed necessary to determine that the antennas meet all the requirements of this specification and the contract, and shall consist of at least the following tests conducted in the order stated:

First - The Service Condition Tests
Second - The Mechanical Stress Test
Third - The Insulation Test
Fourth - The Voltage Standing Wave Ratio Test

4.3.3 Design data. The contractor shall submit with the preproduction models complete design data including manufacturing drawings in accordance with applicable requirements of specification SR-190.

4.3.4 Model acceptance. Acceptance of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No additional antennas shall be delivered prior to the approval of the preproduction models. Prefabrication of any antennas prior to the approval of the preproduction antennas is at the contractor's own risk. The approved preproduction antennas will be returned to the contractor for his use in the fabrication and testing of the antennas to be submitted for acceptance under the contract. The preproduction antennas shall not be considered as equipments under the contract, however, they may be reworked by the contractor and submitted for acceptance as production equipment.

4.4 Inspection tests. Inspection tests shall be conducted on all antennas submitted for acceptance under contract and on all antennas furnished by the contractor with aircraft or other equipment under contract. Inspection tests shall be conducted at the manufacturer's plant under the supervision of the Naval Inspector, unless otherwise

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specified in the contract. Contractors not having laboratory testing facilities satisfactory to the Naval Inspector shall engage the services of a commercial testing laboratory acceptable to the Procuring Service. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification and signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.4.1 Scope of tests. Inspection tests shall consist of the following tests:

(a) Individual tests. Each antenna shall be given the following tests:

The Insulation Test
The Voltage Standing Wave Ratio Test
The Visual Inspection Test

(b) Sample tests. Sample tests shall be conducted on one antenna selected from each lot of fifty or portion thereof submitted for inspection. Samples shall be selected by the inspector and shall first have passed the individual tests. Sample tests shall be more extensive than the individual tests and shall include any of the tests listed under preproduction tests which are deemed necessary by the inspector in addition to the following tests conducted in the order stated.

First - The Mechanical Stress Test
Second - The Insulation Test
Third - The Voltage Standing Wave Ratio Test
Fourth - The Visual Inspection Test

4.5 Test methods and requirements.

4.5.1 Service Condition Tests. The Service Condition Test shall be conducted in accordance with all the requirements of specification MIL-T-5422. The term "satisfactory operation" as used in MIL-T-5422 shall, for this antenna, consist of the Insulation Test of paragraph 4.5.3 and the measurement of voltage standing wave ratio at 240, 300, and 360 megacycles per second. Variation in voltage standing wave ratio of more than 50 percent at any one frequency during the progress of the service condition tests shall be cause for rejection.

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4.5.2 Mechanical Stress Test. The antenna shall be fastened securely in a fixture by means of 3/8 inch bolts and loaded laterally. A uniformly distributed load shall be applied in increments until a static load of 6.9 pounds per square inch has been attained. No yielding of any structural part nor impairment of electrical performance shall result.

4.5.3 Insulation Test. The resistance of the insulation between the stub and the sleeve of the completely assembled antenna shall be measured in accordance with the best commercial practice using a self-contained and self-powered insulation resistance test set having a 500 volt generator and a scale range of 0-1000 megohms with an additional scale marking for infinite resistance. The d-c resistance shall be not less than 50 megohms.

4.5.4 Radiation Test. The Radiation Test shall consist of connecting the antenna through a fifty foot length of RG-21/U cable to a monitored signal source tuned to one frequency within the frequency range of the antenna and observing the relative level of energy radiated from the antenna as indicated on the output meter of an r-f receiving device tuned to the same frequency as the signal source. All conditions of tests shall be identical for each antenna tested. The relative level of radiation shall not vary more than .5 db from an average level determined from the approved preproduction models under identical conditions and methods of test.

4.5.5 Voltage Standing Wave Ratio Tests. The voltage standing wave ratio shall be determined in accordance with the best engineering practice when the antenna is mounted centrally on the convex surface of a semi-cylindrical ground plane of 3.5 feet radius and 10 feet length. The voltage standing wave ratio, measured at or corrected to the antenna connector, shall not be greater than 2.0 on a 50 ohm coaxial transmission line for the frequency range from 225 to 400 megacycles per second.

(a) On preproduction models. Measurements shall be made at 200 mc/s, 215 mc/s, 225 mc/s, and at 20 mc/s intervals from 240 mc/s to 420 mc/s, inclusive.

(b) On inspection samples measurements shall be made at 240 mc/s, 300 mc/s, and 360 mc/s.

4.5.6 Visual Inspection Test. The antenna shall be examined visually to determine that it is properly marked, that its dimensions and construction are as specified, that the workmanship is satisfactory, and that it has not yielded or deteriorated physically under any of the other tests.

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5. PREPARATION FOR DELIVERY

5.1 General. The antennas shall be protected from corrosion or deterioration during shipment and storage. Each antenna shall be packaged, packed and marked for equipment in accordance with the requirements of specification MIL-P-6064. Group XI, Type I packing method shall be used. Shipment marking shall be in accordance with the requirements of specification MIL-STD-129.

6. NOTES.

6.1 Use. The AT-141A/ARC antenna is intended for use with airborne receiving and transmitting equipment in the frequency range from 225 to 400 megacycles per second. A 50-ohm r-f transmission line (such as RG-8/U) is required between the equipment and the antenna.

6.2 This specification is not intended to be restrictive with respect to the details of construction, except where such details are specified. Alternate forms of construction will be considered, provided that the contractor submits to the Bureau of Aeronautics for approval, a clear description of the points of difference and further provided that all performance requirements of this specification are met.

6.3 The right is reserved to reject an materials which have not been subjected to the required tests and found satisfactory.

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DOCUMENT IDENTIFIER AND TITLE
MIL-A-7879A(AS)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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